

To: Data Science Track

From: Abdullahi Ahmad, Data Analytics Intern

Project Phase: Week 6 – Cross-Track Integration & Advanced Development

Subject: High-Weight Feature Recommendations for No-Show Classification Model

1. Integration Objective

The purpose of this handoff is to provide the Data Science track with validated, evidence-based analytical findings from the Week 6 Exploratory Data Analysis (EDA). These insights should be utilized to inform feature selection and weighting for the baseline classification model predicting patient no-shows.

2. Key Analytical Findings

Based on the advanced multivariate analysis of 5,000 scheduled visits, the following operational bottlenecks have been identified:

The Lead Time Danger Zone: The clinic is operating with a critical ~48.5% no-show rate. There is a direct mathematical correlation between advanced booking lead times and attendance. Patients who fail to attend book significantly further in advance (averaging ~34 days) compared to attendees (averaging ~24 days).

Reminder Channel Inefficiencies: A significant volume of missed appointments correlates directly with the "None" reminder category.

Demographic Variances: Attendance behaviors fluctuate significantly across age brackets when crossed with specific reminder channels, indicating that uniform communication strategies are failing certain patient segments.

3. Strategic Recommendations for ML Modeling

To ensure the predictive model is highly accurate and aligned with the actual operational data, the Data Analytics track formally recommends that the Data Science track heavily weights the following features in their algorithm:

booking_lead_days

reminder_channel

age_group

Please confirm if these variables are currently registering as high-weight features in your baseline classification model so we can validate our cross-track findings.